
DEVOXX™
Raphaël Semeteys
2025 October 7
GenAI & Beyond
Tools-in-Action 30 min.
BEGINNER
The Yoga of Image Generation
with Stable Diffusion & ComfyUI

Raphaël Semeteys

Head of DevRel
Senior Architect, Opensource Expert

- **Worldline** since 1999
- Based in Paris
- raphiki.github.io
- **Yoga Teacher**



Contact

@RaphaelSemeteys

dev.to/raphiki/

linkedin.com/in/raphaelsemeteys/

Who are we?

We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 33+ billion
transactions per year



€250M spent in products
and technology per year



Handling 150+
payment methods

DEVOXX™

WORLDLINE

Use Case

Generate images of Yoga poses

- Photo not always suitable
- No images from the Internet
- Poses must be precise



How could
Generative AI help?



From German Labs to London Startup

- **Collaboration** between Labs and Companies
  - Latent Diffusion model in 2021
  - Participation of Runway and EleutherAI
- **Stability AI**
  - Donation of computing power to the project
  - Hiring most of the initial team
  - Now maintainer of Stable Diffusion models
- Vibrant **Community** (models, tools, tips)



runway



stability.ai

"Open" licenses

- Responsible AI: OpenRAIL
- Version 3.5: 1M+ revenue companies must pay



From German Labs to London Startup

- **Collaboration** between labs and companies
  - Latent
  - Using
  - Partici
- **Stability AI**
  - Donation of computing power to the project
  - Hiring most of the initial investigators
  - Now maintainer of Stable Diffusion modelsEleutherAI
- Vibrant Contributing Communities (models, tools, tips)



More on this topic tomorrow!

2pm, room 6

From OpenAI to OpenSource AI



runway

stability.ai

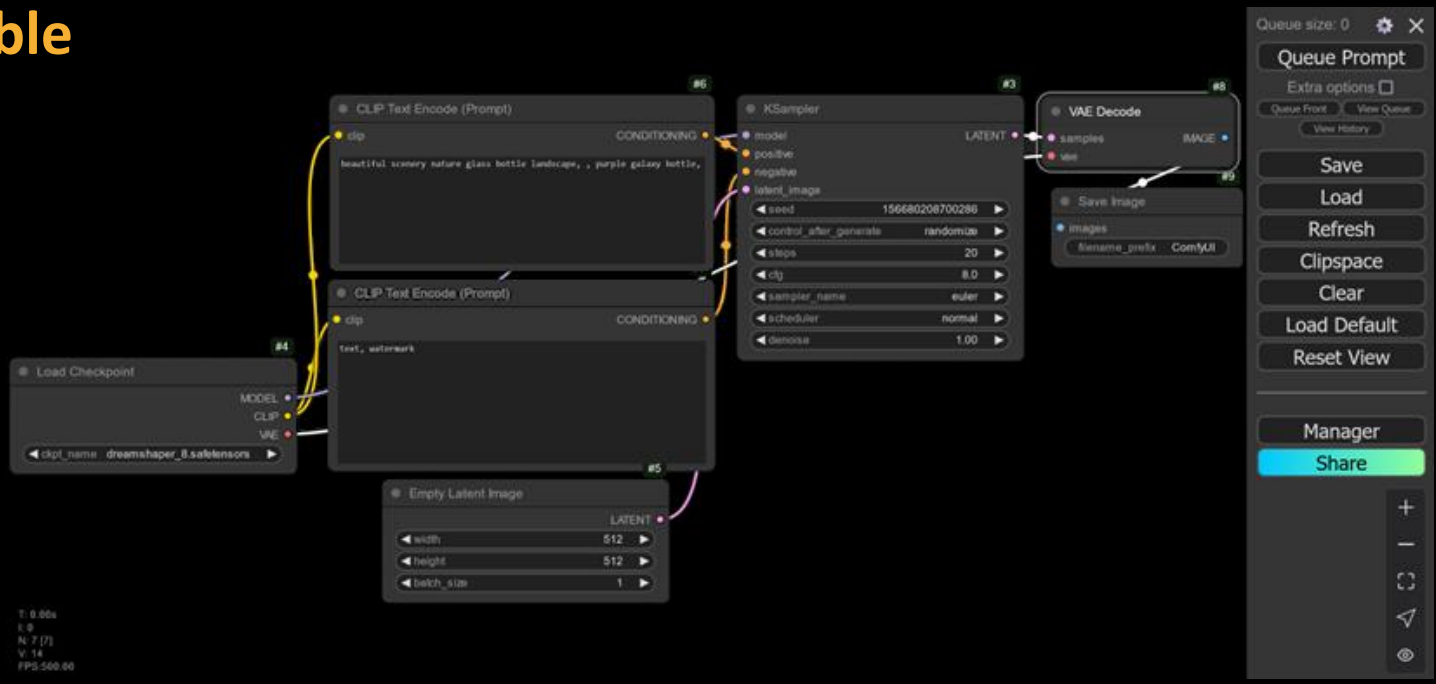
"Open" licenses

- Responsible AI: OpenRAIL
- Version 3.5: 1M+ revenue companies must pay

Comfy

Local GUI for Creative Workflows

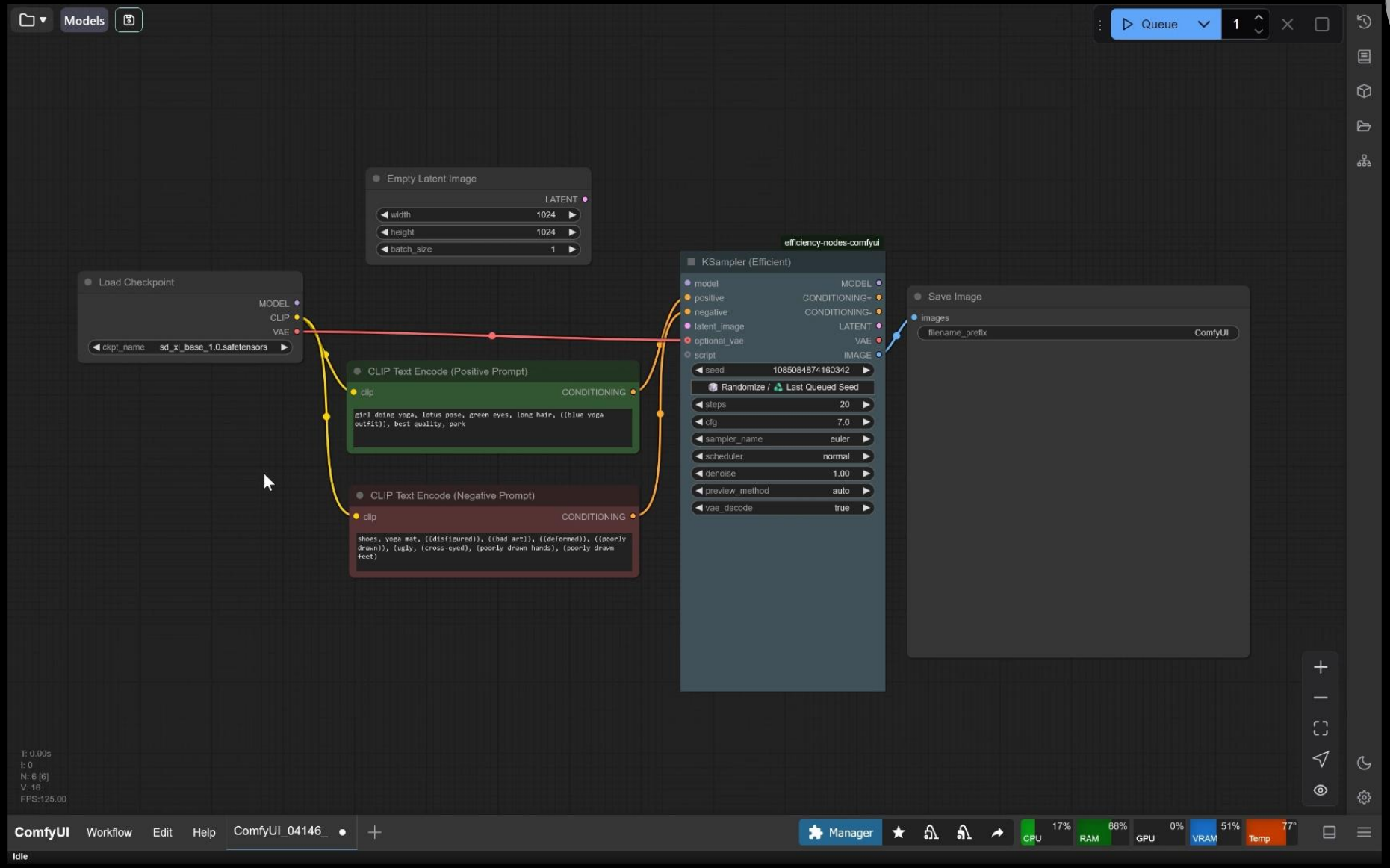
- **Intuitive, modular and customizable**
- **Flexible node-based workflows**
  - Text-to-Image Generation
  - Image-to-Image Processing
- **Community aspect**
  - GPLv3 license
  - Custom nodes



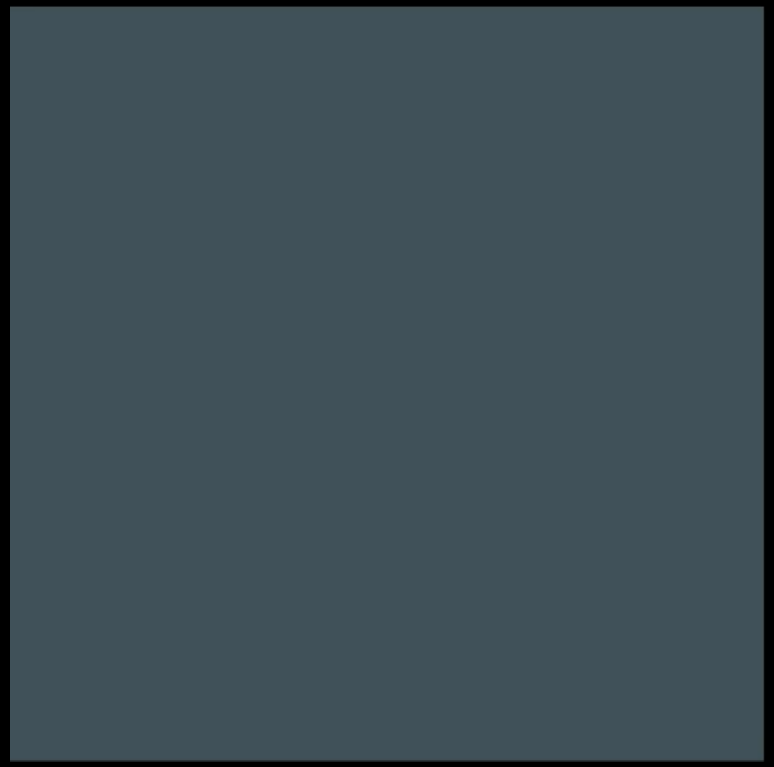
Let's start with a simple demo

Generate an image of a
woman doing yoga

Demo – Text-to-Image (T2I)



How does Stable Diffusion work?



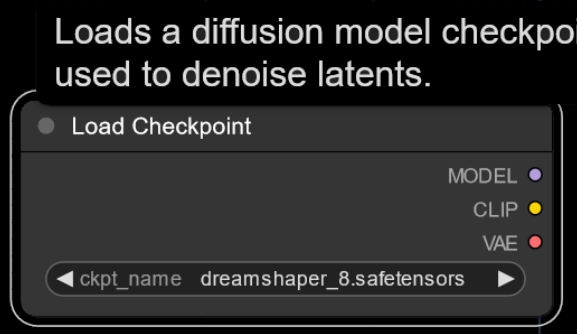
- **Starts with a random noise**: unrecognizable image
- **Refines step by step**: gradually suppresses noise by adding details
- **Learns from real images**: injects patterns from trained images
- **Is text-driven**: prompts like "Woman doing yoga in a park“
- **Produces a final image**: detailed and aligned with the prompt

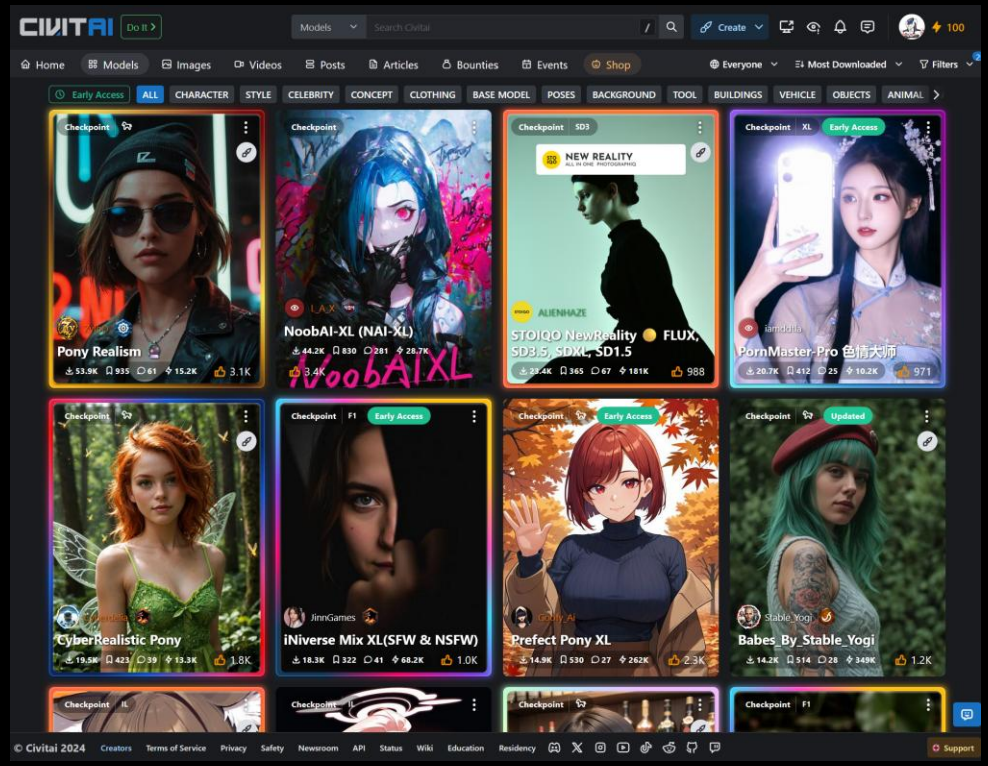
Latent Space

Models

- **Most used Stability.ai models**
  - SD 1.5
  - SDXL
- **Fine-tuned Models**
  - Provided by dynamic communities
  - Hugging Face, civitai.com

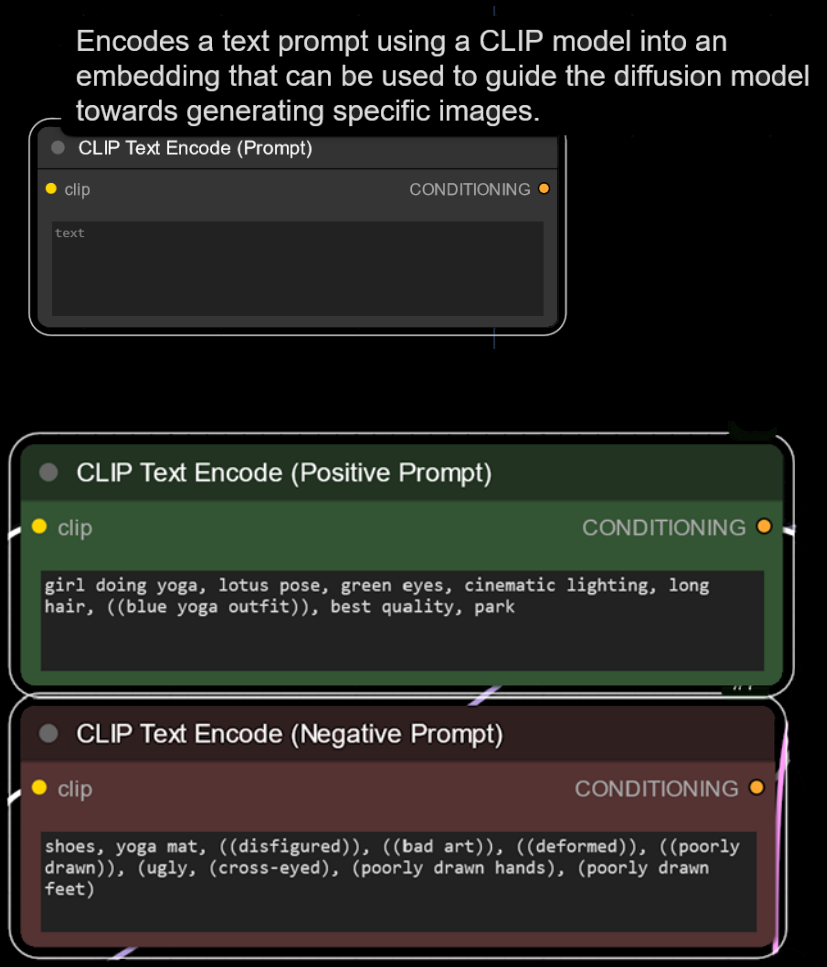
CHARACTER STYLE CELEBRITY CONCEPT CLOTHING BASE MODEL POSES
BACKGROUND TOOL BUILDINGS VEHICLE OBJECTS ANIMAL ASSETS ACTION





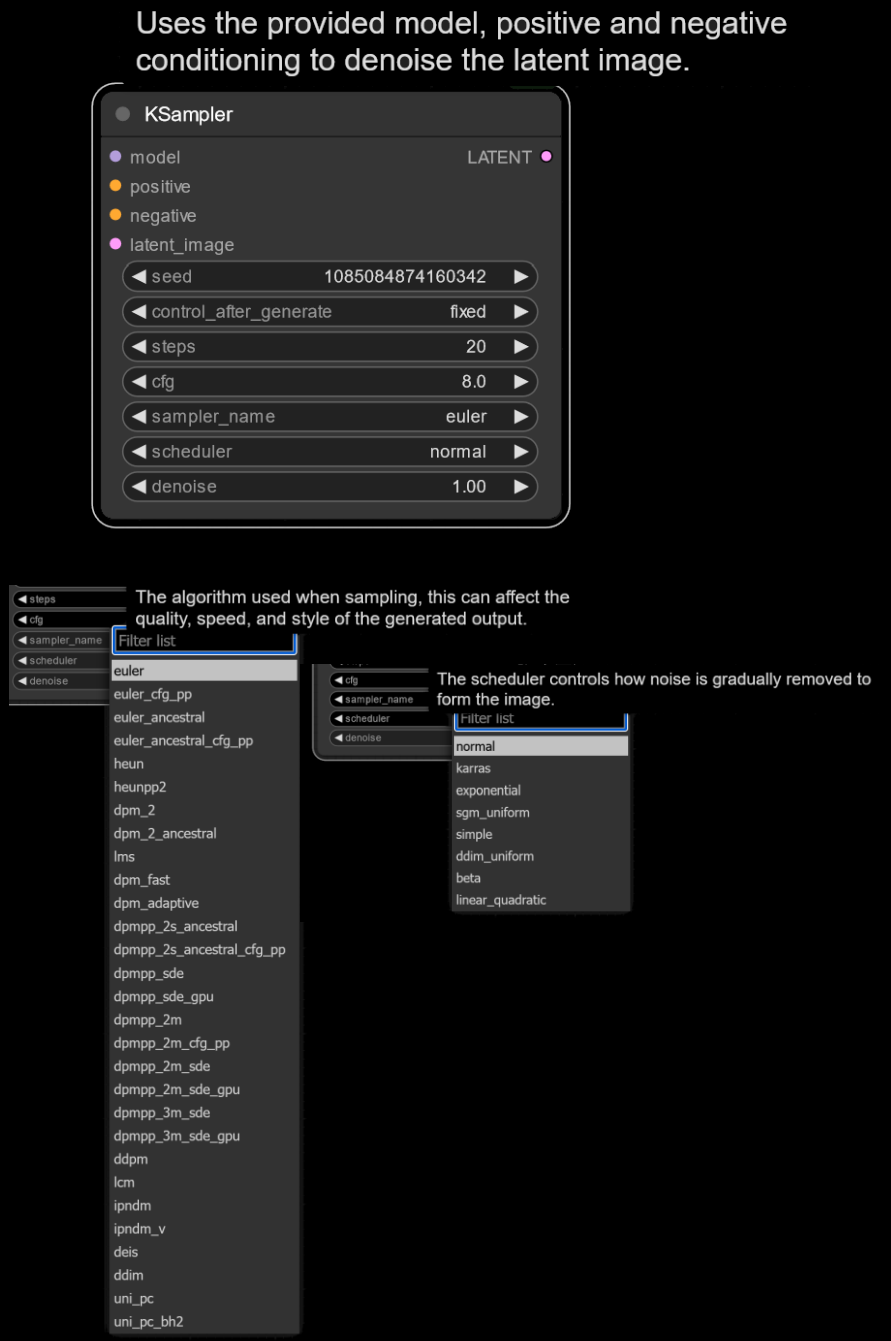
Prompts

- **CLIP Model** (Contrastive Language–Image Pretraining)
  - Developed by OpenAI in 2021
  - Understands the prompt
  - Guides generation
- **Positive and negative Prompts**
- **Textual or short Syntax**



Denoising process

- **Samplers**
  - Algorithms guiding Iterative Image Generation
  - Differ in speed and quality
- **Schedulers**
  - Control how noise is eliminated at each stage
  - Also impact speed and quality
- **Other settings**
  - # steps, CFG (prompt impact), % denoising
- **Seed**
  - Random element used to create the initial noise
  - Fixing it allows to check impact of other parameters



We can tweak the prompt



long hair,
((blue yoga outfit))



long red hair,
((pink yoga outfit))

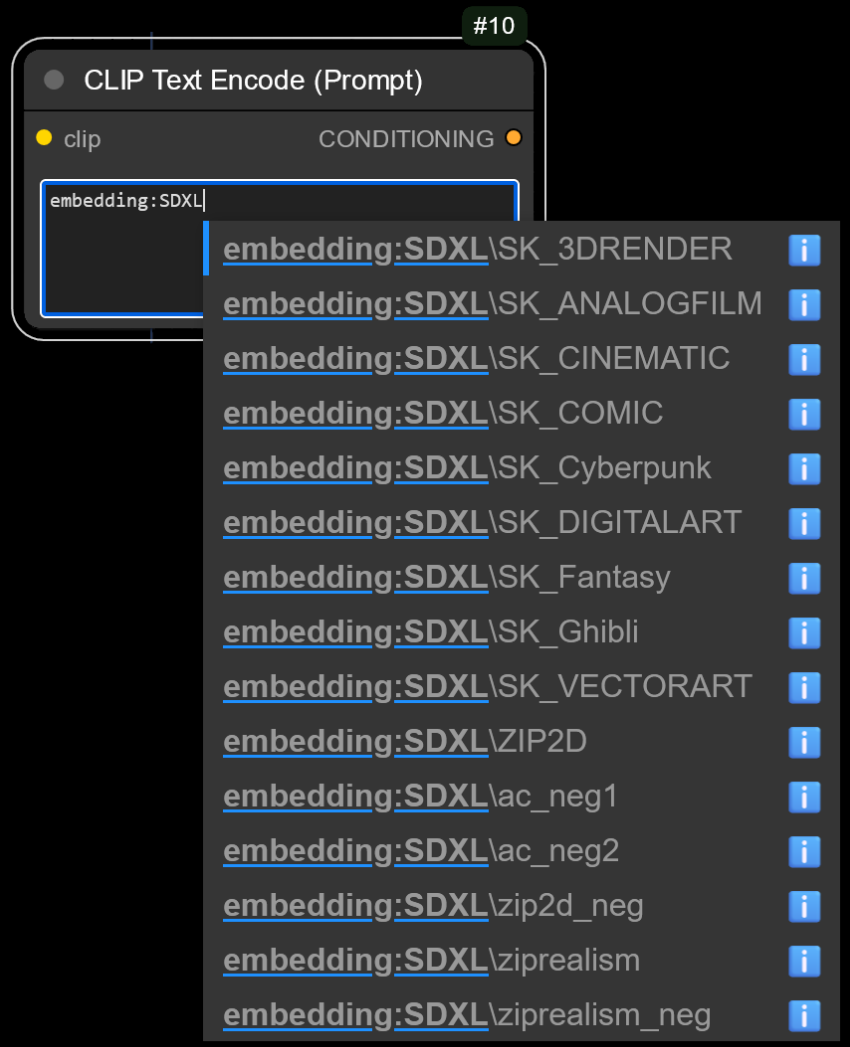


mid long dark hair,
long white t-shirt,
((blue yoga pants))

Textual Inversions (Embeddings)

- **Vector Representations of Text**
  - Pre-defined *instructions* for image generation
  - Style, theme, texture, pose, character, etc.

- **Files containing additional concepts**
  - Injected into the prompt
  - The community provides many presets
  - Must be aligned with the Model's version



Textual Inversions (Embeddings)


Initial


3D Render

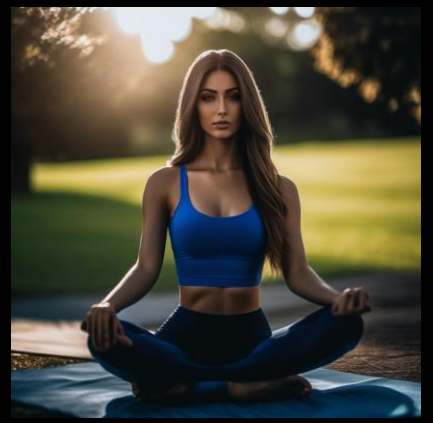

Comic


Fantasy


Ghibli


Vector Art


Analog Film


Cinematic


Cyberpunk


Digital Art

We can tweak the style
But no control on the pose...



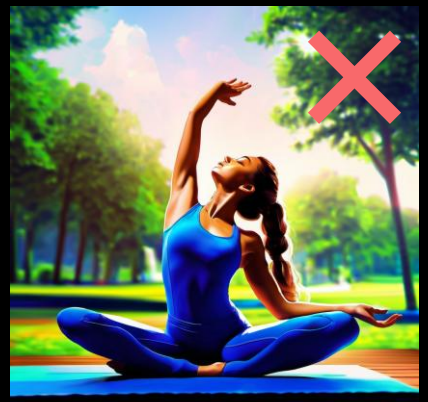
Lotus
pose



Tree
pose



Camel
pose



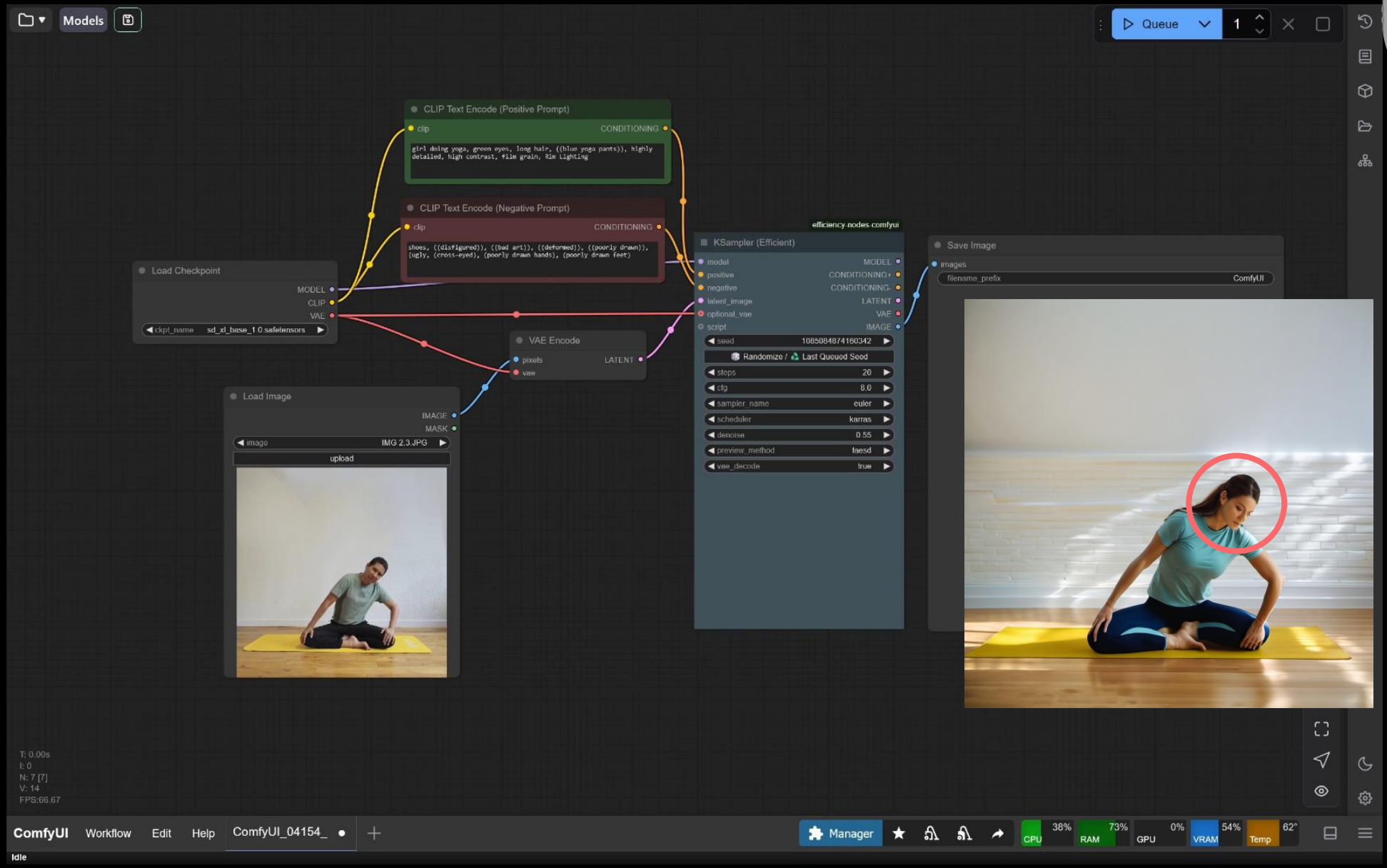
Shoulder-stand
pose

Text-to-Image generation is not enough!

Let's move on to Image-to-Image generation

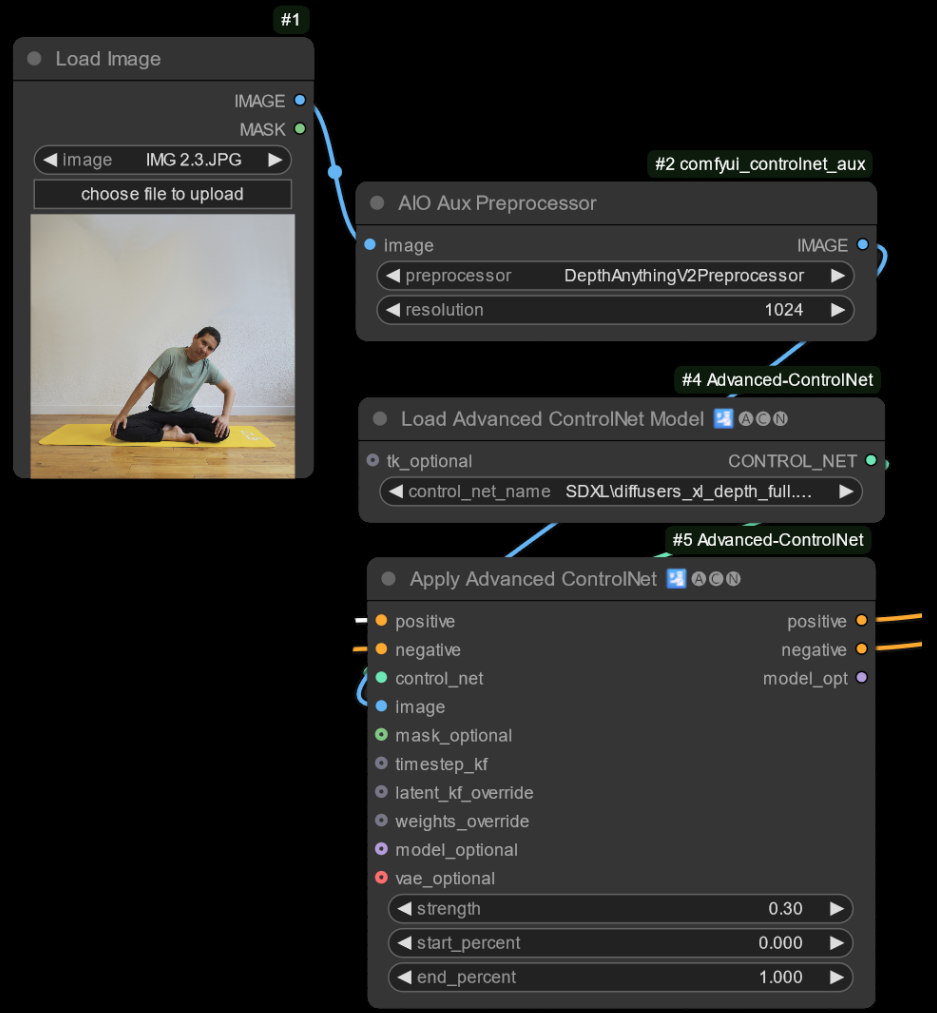
Generate an image of a woman doing yoga **based on an existing image**

Demo – Image-to-Image (121)

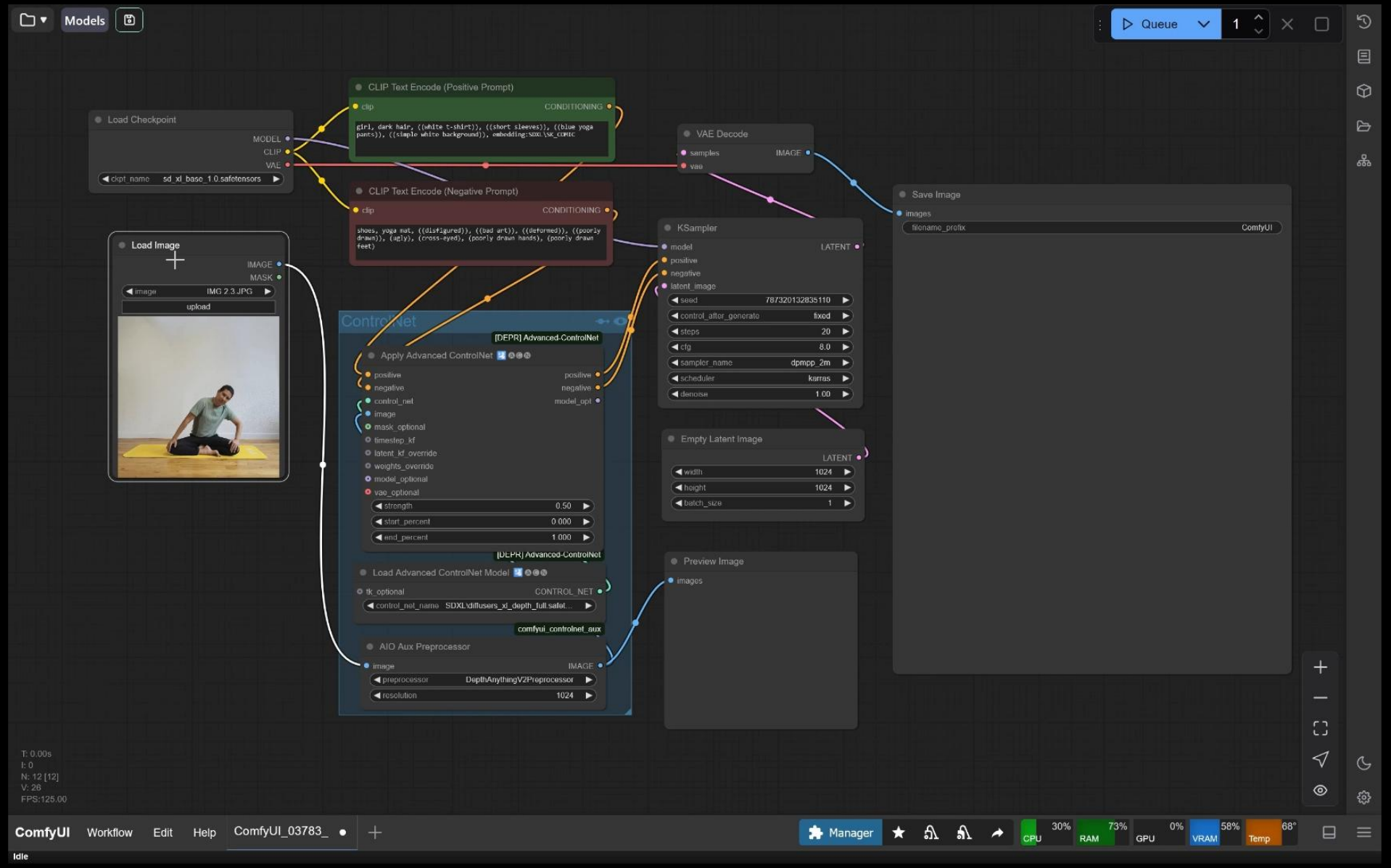


ControlNets

- **Specialized Neural Networks**
  - Additional control and guidance
  - Transfer only certain features of an image
- **Control methods**
  - Structure: pose, edges, segmentation, depth
  - Texture and detail: sketching, stylizing from edges
  - Content and layout: bounding boxes, masks
  - Abstract and style: color maps, texture fields



Demo – I2I with ControlNet



Examples of ControlNet Preprocessors



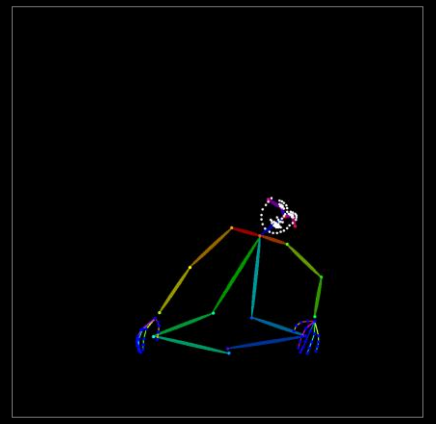
Initial Image



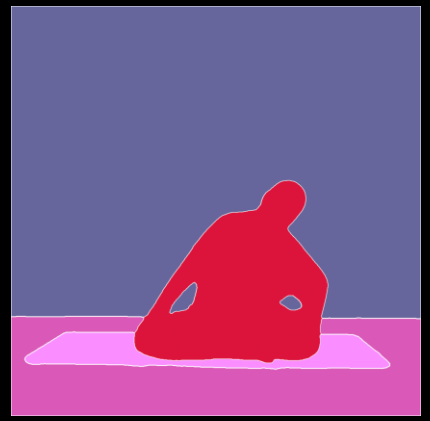
Line Art



Color Map



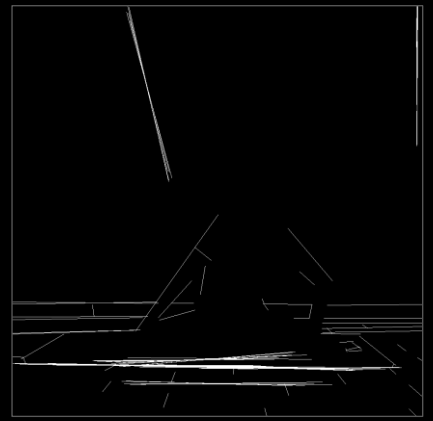
Open Pose



Segmentation



Depth Map



Straight Lines



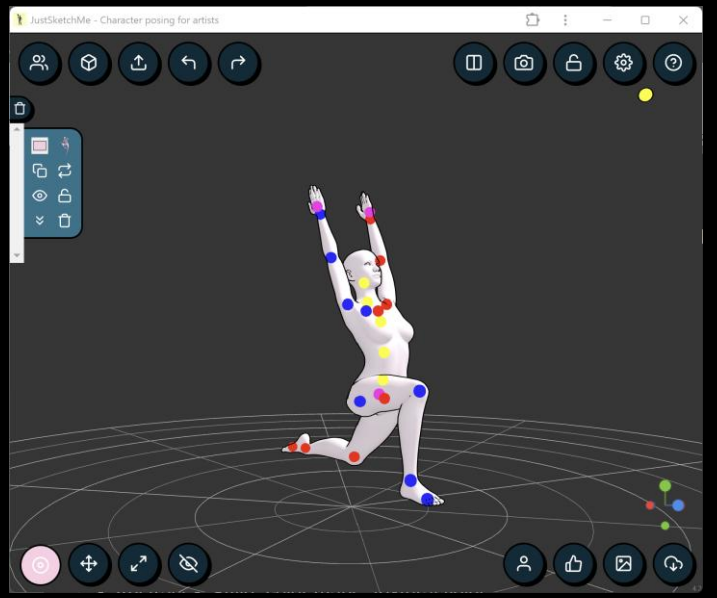
Scribble

More abstract input images

3D poses exported as an images

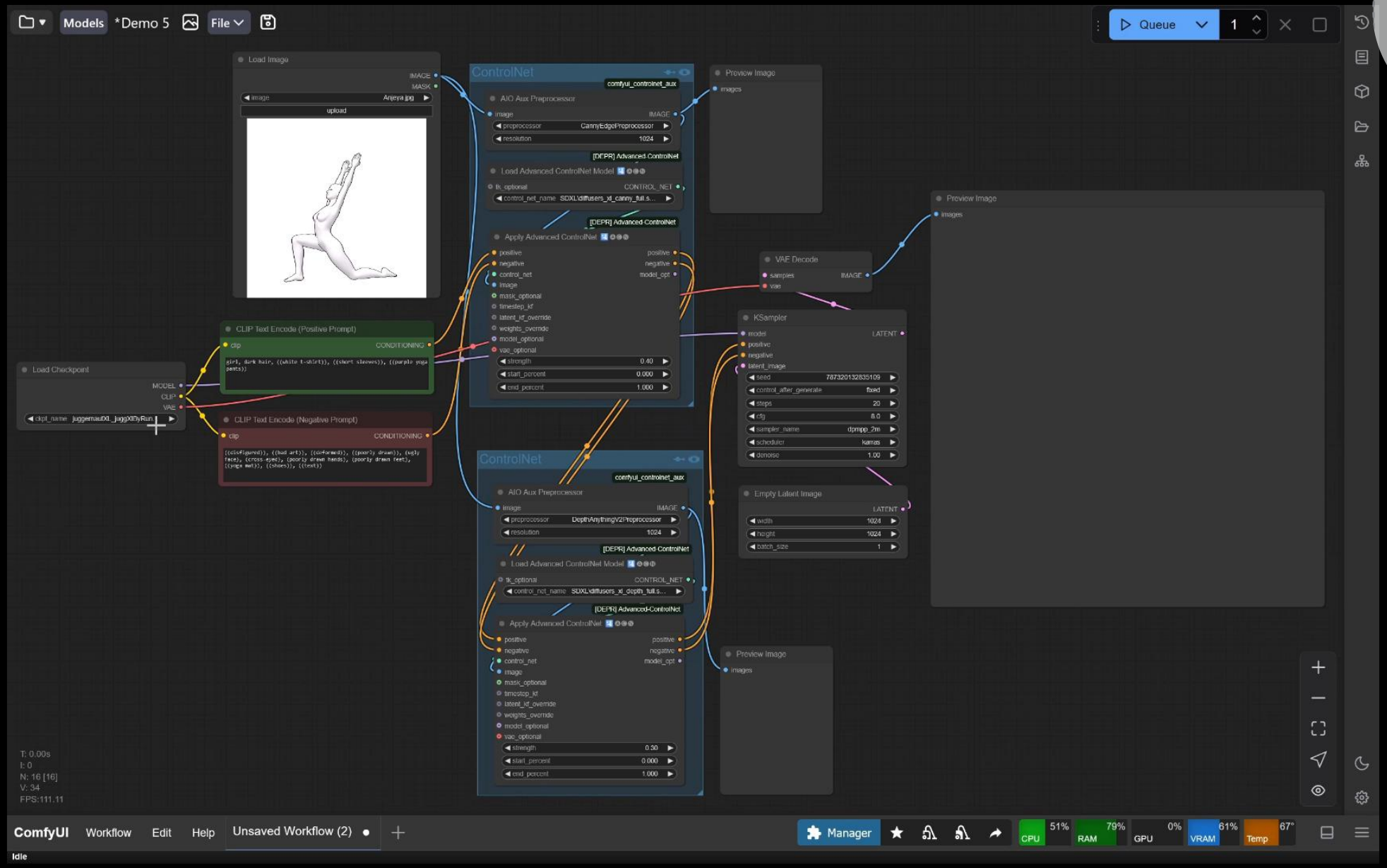
- Using the JustSketchMe tool (webapp & PWA)
- Modeling based on my own knowledge
- Multiple viewing angles





JustSketchMe

Demo – T2I with 2 ControlNets

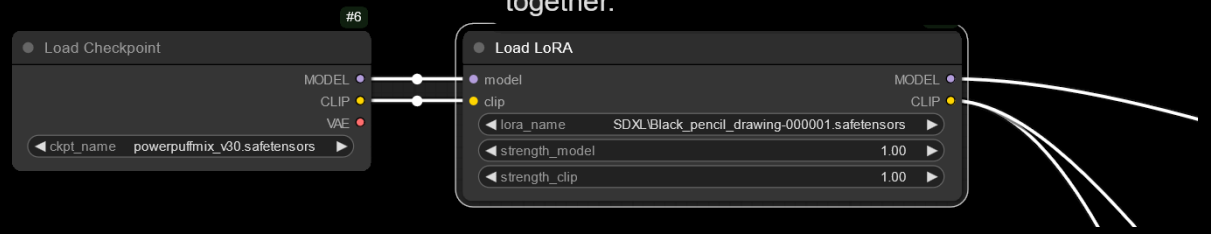


How to get more consistency within a sequence?

Generate a series of images with the same **identity**

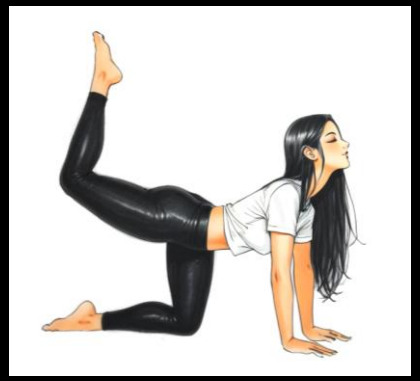
LoRA

LoRAs are used to modify diffusion and CLIP models, altering the way in which latents are denoised such as applying styles. Multiple LoRA nodes can be linked together.

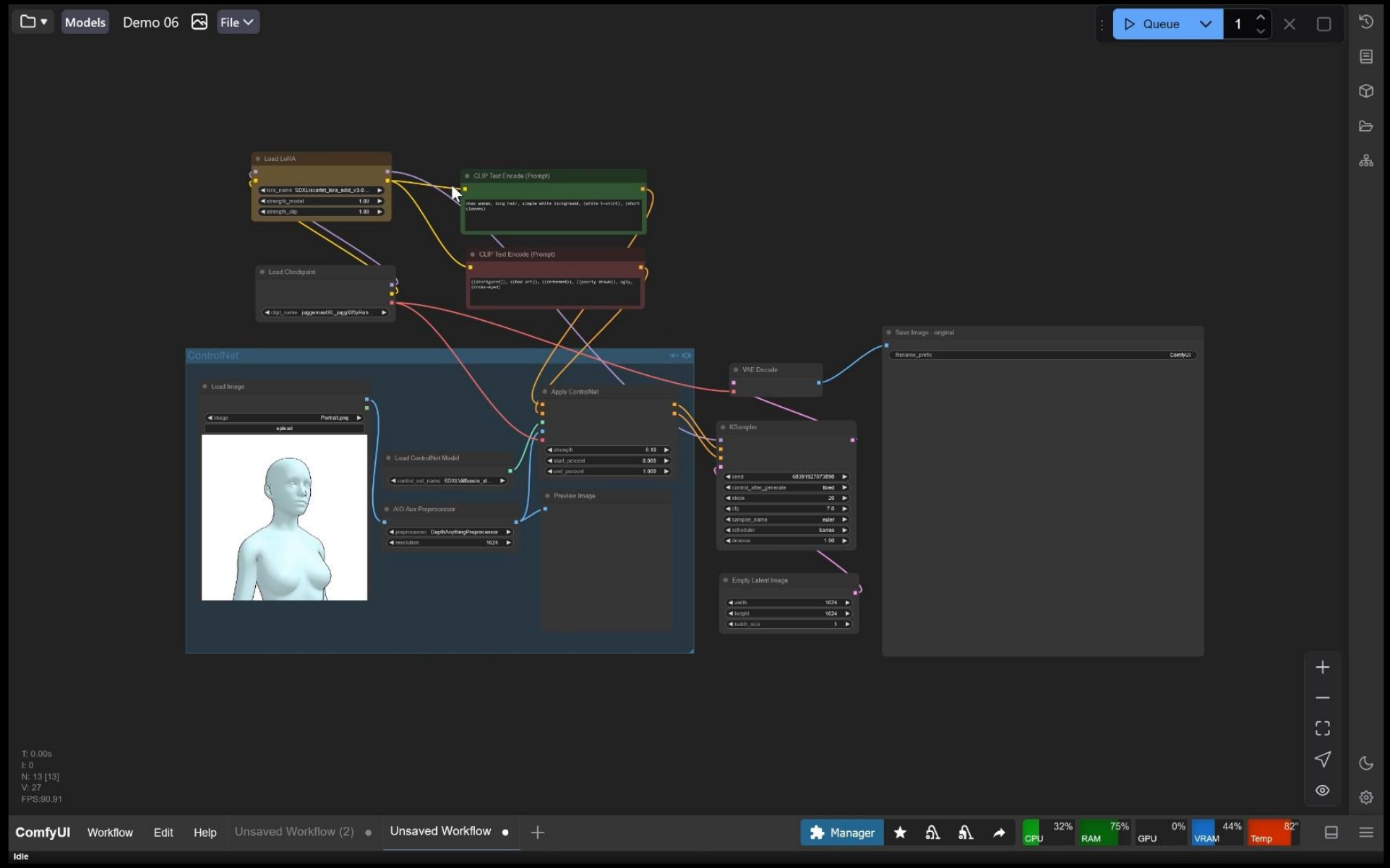


- **Low-Rank Adaptation**
  - Slight adaptation of the basic model
  - Like a smaller “patch” of the model at runtime
- **Usage**
  - Triggered by keywords in the prompt
  - Numerous LoRAs provided by the community
  - Styles, poses, concepts, or characters

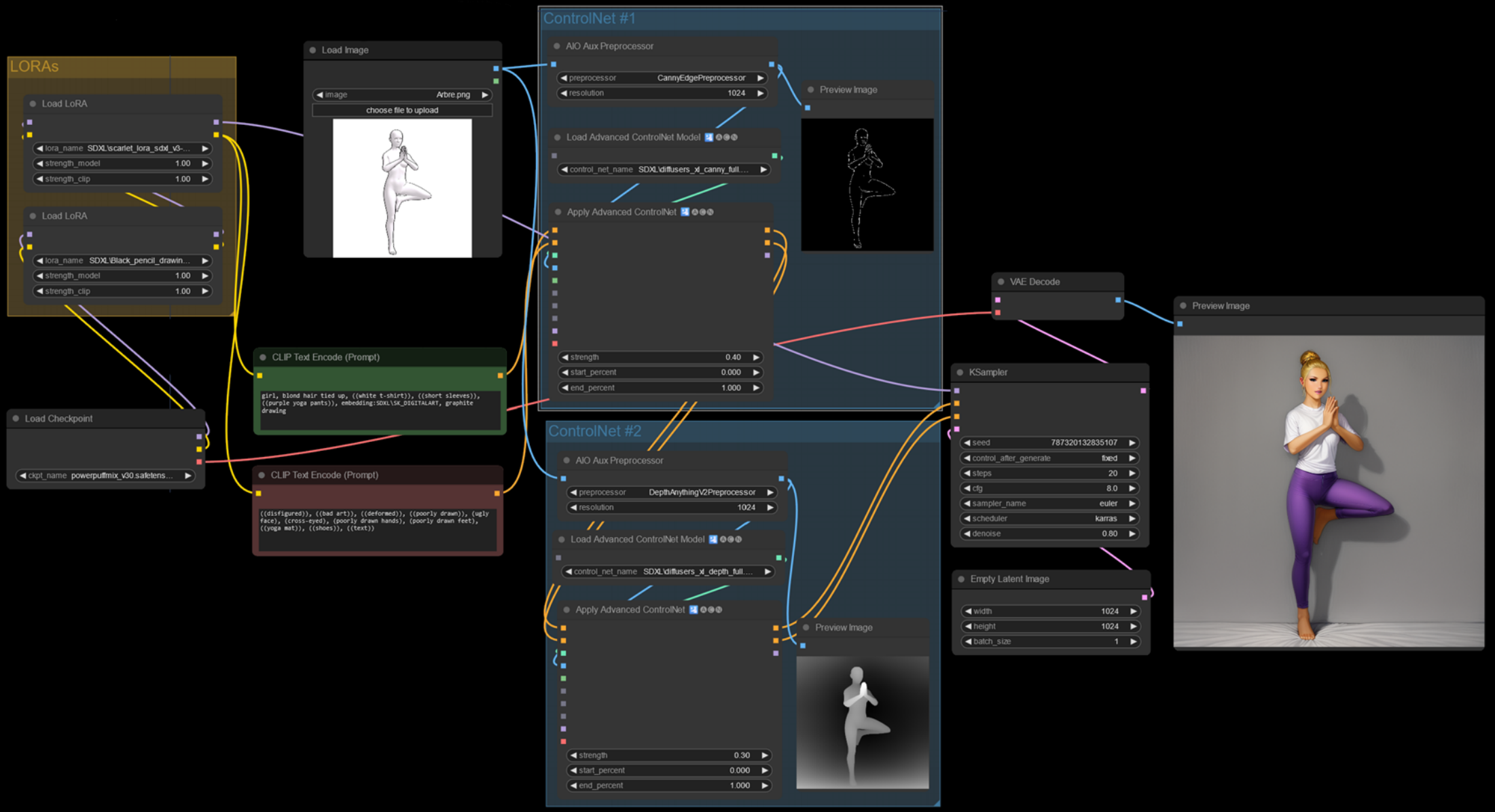




Demo – ControlNet + LoRA



Embeddings + 2 ControlNets + LoRA



Embeddings + 2 ControlNets + LoRA



Embeddings + 2 ControlNets + LoRA



My own Haj3r LoRA

Dreambooth

- Based on SDXL
- 28 input images
- 2 epochs
- Google Colab Notebook
- 1:30 (remote)





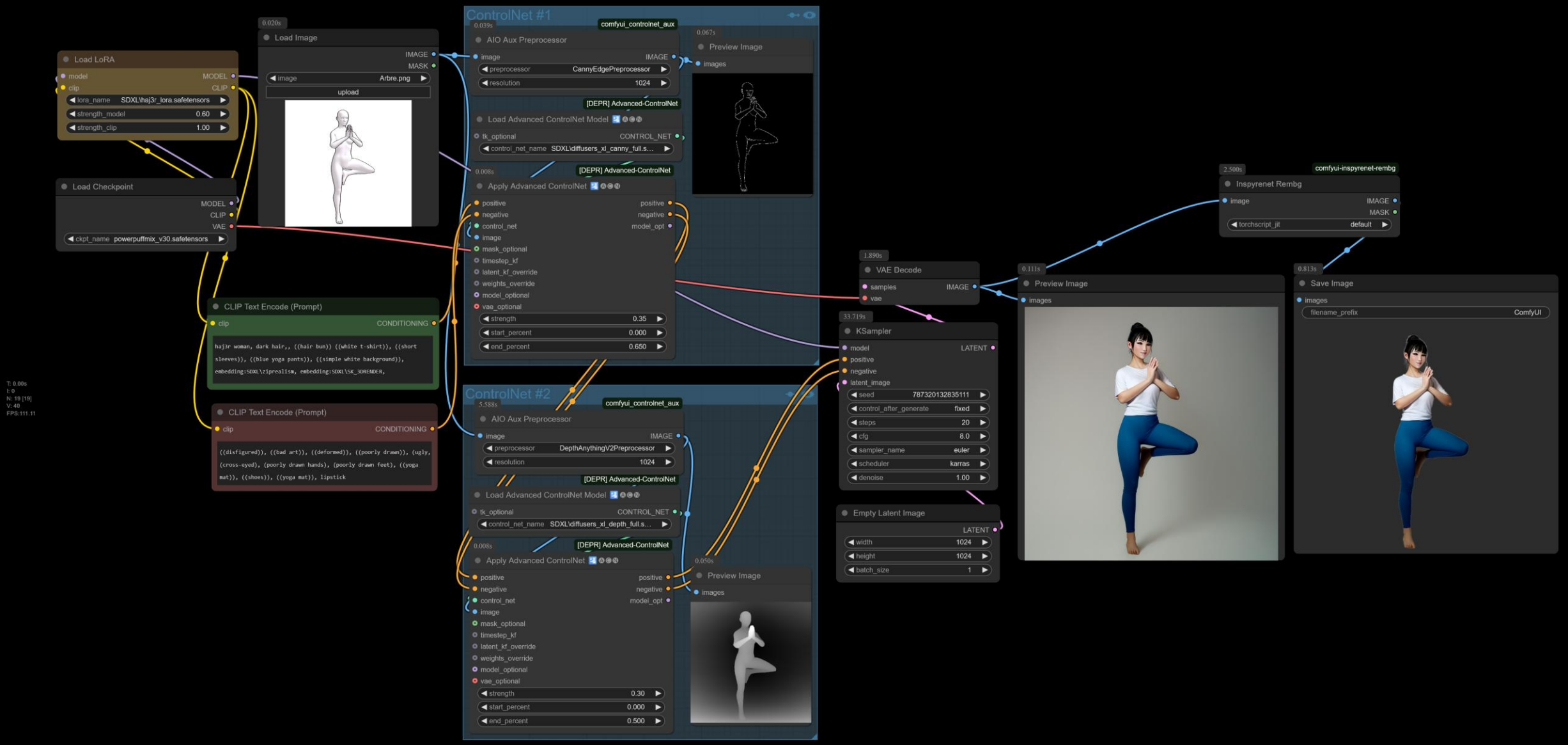
Kohya_ss

- Based on PowerPuffMix
- 15 input images
- 20 epochs
- 3:30 (local)
- Embeddings





t2i + Embeddings + ControlNets + Haj3r LoRA

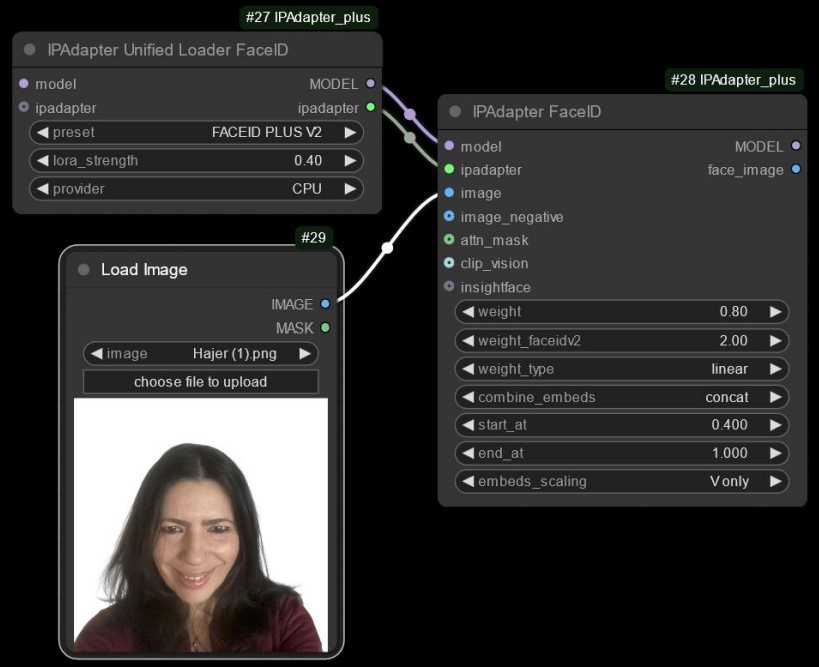


t2i + Embeddings + ControlNets + Haj3r LoRA

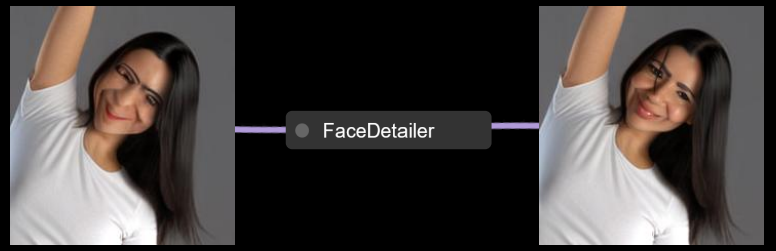


FaceID + FaceDetailer

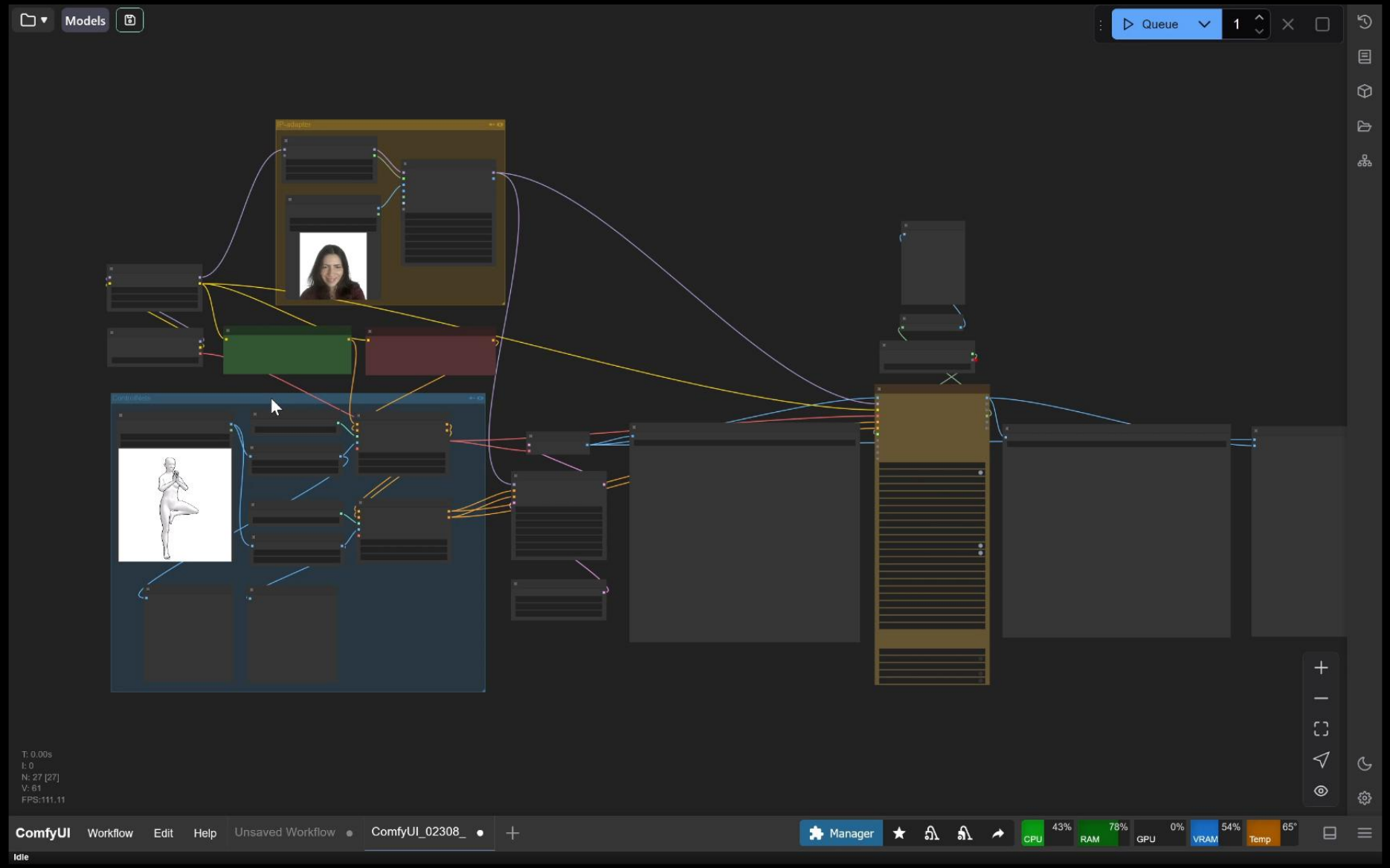
- **Image Prompt Adapter**
  - Kind of one-image LoRA
  - FaceID: facial recognition IP Adapter



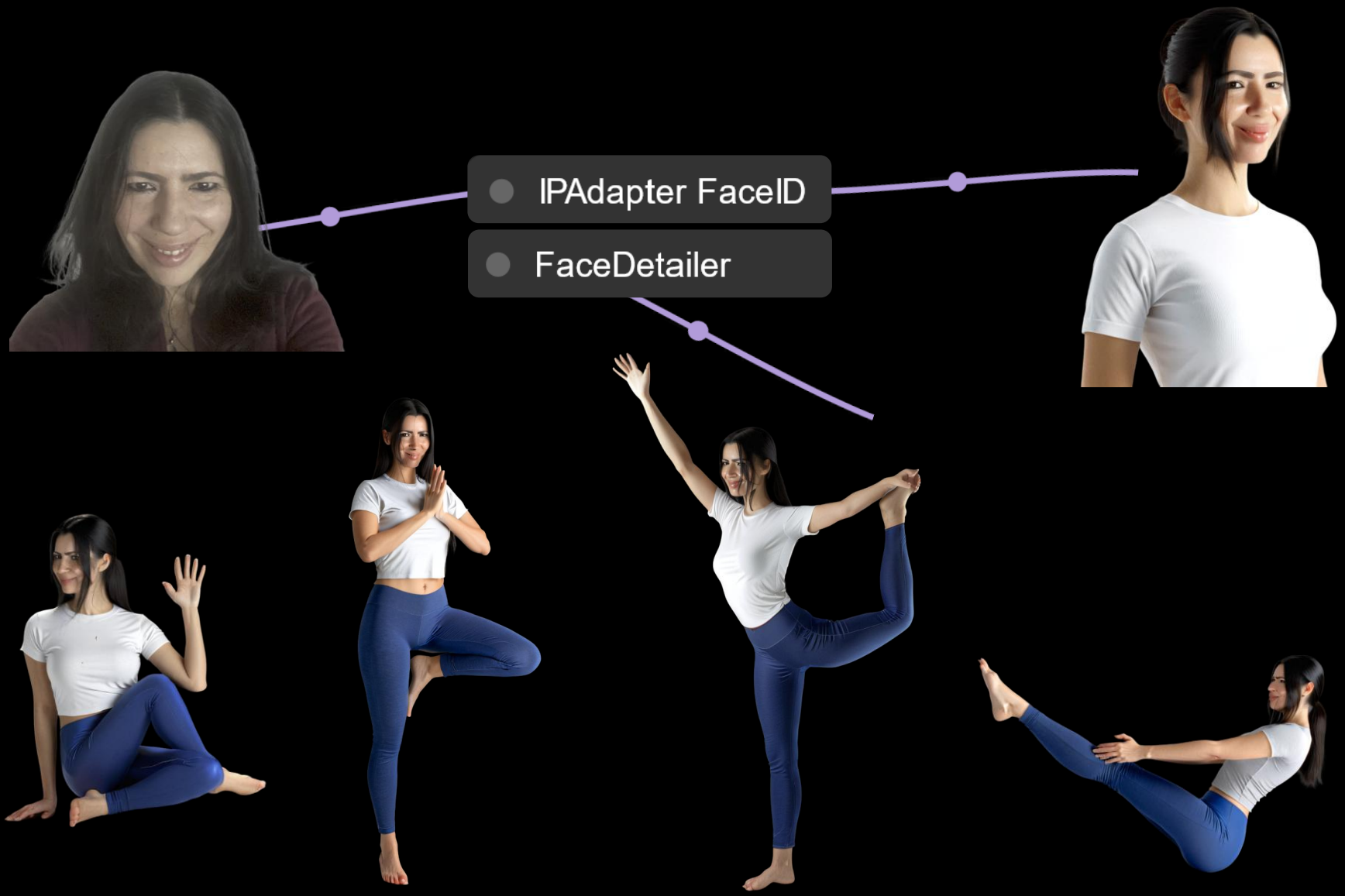
- **FaceDetailer**
  - Post-processing AI model
  - Facial enhancement (eyes, nose, lips, expression)



Demo – ControlNets + FaceID + FaceDetailer



ControlNets + FaceID + FaceDetailer



Easier to switch Models

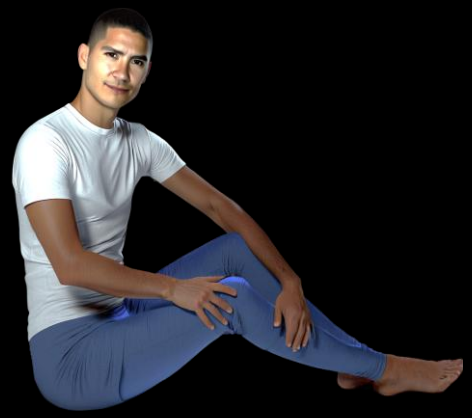
Cheyenne v2



Easier to switch Personas













Let's Summarize



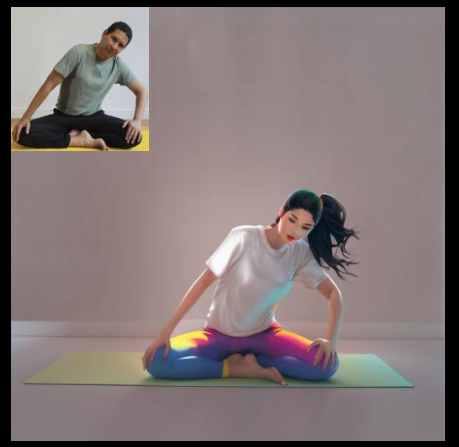
t2i



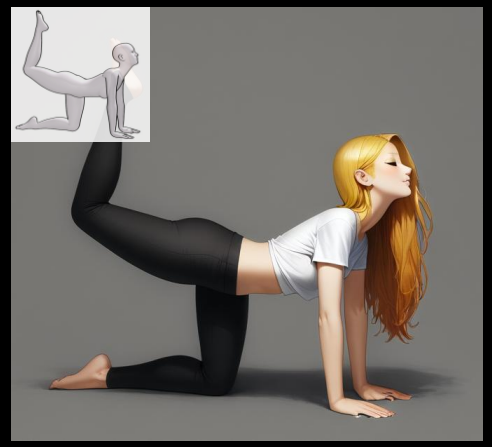
t2i + embeddings



t2i + i2i



t2i + embeddings + ControlNet



t2i + embeddings + ControlNets + LoRA



t2i + embeddings + ControlNets + Haj3r LoRA



t2i + embeddings + ControlNets + FaceID + FaceDetailer

Implemented use case

Accurate & consistent images

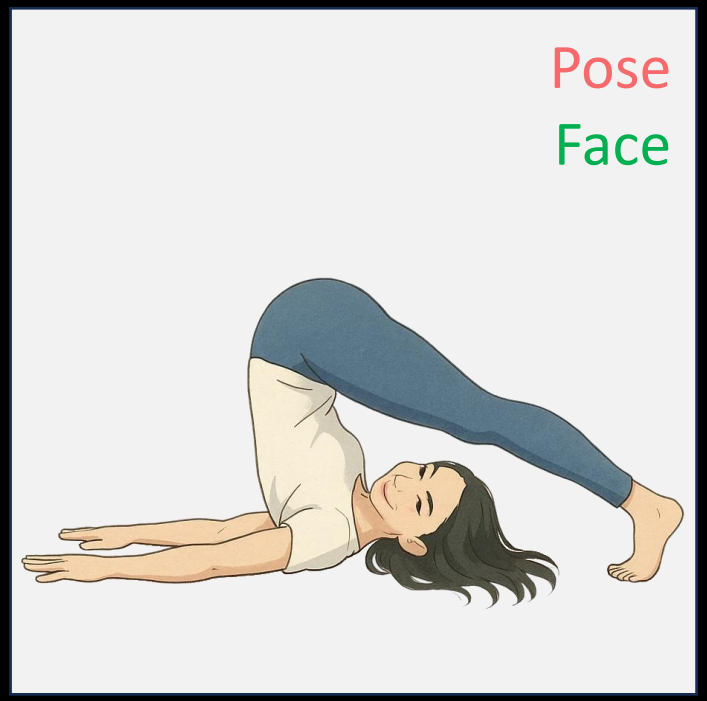
Science & Art

Numerous nodes & parameters

Icing on the cake: more **inclusivity**



What about New Multimodal Models?



OpenAI GPT-Image-1



Google Nano Banana



Black Forest Labs Flux.1

Less control on both pose and face (yet)

What about New Multimodal Models?



Very useful editing features

Next step... Video?

Comfy

Wan



The Yoga of Image Generation



स्थिरसुखमासनम्

Yoga Sūtra II.46

The Pose should be Stable and Comfortable



Thank You!

raphiki.github.io

Your Feeback

DEVOXX

WORLDLINE